



Tables with mean and variance

Single-Task Baselines and Distillation Metrics

epochs = 10, KD without teacher annealing, T = 1, full datasets

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
ledgar_Teacher	1	8.00 / 0.000000	0.8252 / 0.000000	0.8214 / 0.000000	0.8833 / 0.000000	label_15 (1.00)	label_8 (0.00)	24.50 smp/s / 0.0000
ledgar_Baseline	3	9.67 / 0.333333	0.5579 / 0.000001	0.7911 / 0.000004	0.8614 / 0.000007	label_14 (1.00)	label_8 (0.00)	348.53 smp/s / 41.5733
ledgar_Single_task_KD_Student	3	9.67 / 0.333333	0.7487 / 0.026168	0.7822 / 0.000105	0.8600 / 0.000004	label_14 (1.00) [S1] / label_93 (0.99) [S2] / label_39 (0.99) [S3]	label_8 (0.00)	342.63 smp/s / 172.9433
ledgar_Mix_05	1	9.00 / 0.000000	0.5589 / 0.000000	0.7926 / 0.000000	0.8630 / 0.000000	label_14 (1.00)	label_8 (0.00)	343.80 smp/s / 0.0000
ledgar_Mix_07	1	9.00 / 0.000000	0.5598 / 0.000000	0.7915 / 0.000000	0.8622 / 0.000000	label_14 (1.00)	label_8 (0.00)	380.50 smp/s / 0.0000
[TF-IDF] ledgar_tfidf	1	10.00 / 0.000000	3.6479 / 0.000000	0.1999 / 0.000000	0.4269 / 0.000000	label_11 (0.95)	label_0 (0.00)	48464.60 smp/s / 0.0000
unfair_tos_Teacher	1	10.00 / 0.000000	0.0243 / 0.000000	0.8335 / 0.000000	0.8117 / 0.000000	label_6 (0.97)	label_2 (0.73)	30.40 smp/s / 0.0000
unfair_tos_Baseline	3	8.00 / 4.000000	0.0233 / 0.000001	0.7764 / 0.000100	0.7421 / 0.000019	label_5 (0.96) [S1,S3] / label_6 (0.97) [S2]	label_2 (0.68) [S1] / label_1 (0.66) [S2] / label_0 (0.68) [S3]	380.97 smp/s / 2816.5733
unfair_tos_Single_task_KD_Student	3	8.67 / 1.333333	0.0278 / 0.000015	0.7781 / 0.000645	0.7443 / 0.000353	label_5 (0.96) [S1,S3] / label_6 (0.97) [S2]	label_0 (0.67) [S1] / label_0 (0.68) [S2] / label_2 (0.65) [S3]	344.07 smp/s / 265.5233
unfair_tos_Mix_05	1	8.00 / 0.000000	0.0233 / 0.000000	0.7805 / 0.000000	0.7418 / 0.000000	label_5 (0.96)	label_0 (0.68)	339.40 smp/s / 0.0000
unfair_tos_Mix_07	1	8.00 / 0.000000	0.0233 / 0.000000	0.7724 / 0.000000	0.7377 / 0.000000	label_5 (0.96)	label_0 (0.67)	345.40 smp/s / 0.0000
[TF-IDF] unfair_tos_tfidf	1	1.00 / 0.000000	0.6859 / 0.000000	0.0284 / 0.000000	0.0285 / 0.000000	label_0 (0.06)	label_5 (0.00)	82059.90 smp/s / 0.0000

Multi-Task Learning Architecture Performance

epochs = 10, KD without teacher annealing, T = 1, full datasets

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
multi_task_model_supervised (System)	3	10.00 / 0.000000	0.4884 / 0.000005	0.7316 / 0.000101	0.7555 / 0.000154	---	---	---
└─ LEDGAR Sub-Split	3	-	0.5604 / 0.000007	0.7929 / 0.000005	0.8626 / 0.000001	label_14 (1.00)	label_8 (0.00)	368.40 smp/s / 53.1200
└─ UNFAIR-ToS Sub-Split	3	-	0.0432 / 0.000005	0.6704 / 0.000323	0.6485 / 0.000562	label_6 (0.87) [S1,S3] / label_5 (0.89) [S2]	label_7 (0.45) [S1] / label_7 (0.52) [S2,S3]	373.10 smp/s / 1214.6800
multi_task_model_kd (System)	3	10.00 / 0.000000	0.6489 / 0.018249	0.7244 / 0.000139	0.7532 / 0.000216	---	---	---
└─ LEDGAR Sub-Split	3	-	0.7459 / 0.024334	0.7819 / 0.000102	0.8601 / 0.000012	label_14 (1.00) [S1] / label_26 (0.99) [S2] / label_93 (0.99) [S3]	label_8 (0.00)	361.90 smp/s / 309.8700
└─ UNFAIR-ToS Sub-Split	3	-	0.0484 / 0.000031	0.6669 / 0.000293	0.6463 / 0.000693	label_5 (0.85) [S1] / label_6 (0.89) [S2] /	label_7 (0.47) [S1,S3] / label_7 (0.54) [S2]	339.40 smp/s / 421.2300

						label_6 (0.88) [S3]		
multi_task_kd_mix_05 (System)	1	10.00 / 0.000000	0.4891 / 0.000000	0.7438 / 0.000000	0.7724 / 0.000000	---	---	---
└─ LEDGAR Sub-Split	1	-	0.5615 / 0.000000	0.7950 / 0.000000	0.8643 / 0.000000	label_14 (1.00)	label_8 (0.00)	347.50 smp/s / 0.0000
└─ UNFAIR-ToS Sub-Split	1	-	0.0416 / 0.000000	0.6927 / 0.000000	0.6805 / 0.000000	label_5 (0.88)	label_7 (0.47)	333.40 smp/s / 0.0000
multi_task_kd_mix_07 (System)	1	10.00 / 0.000000	0.4905 / 0.000000	0.7385 / 0.000000	0.7708 / 0.000000	---	---	---
└─ LEDGAR Sub-Split	1	-	0.5630 / 0.000000	0.7926 / 0.000000	0.8642 / 0.000000	label_14 (1.00)	label_8 (0.00)	349.70 smp/s / 0.0000
└─ UNFAIR-ToS Sub-Split	1	-	0.0417 / 0.000000	0.6844 / 0.000000	0.6773 / 0.000000	label_5 (0.85)	label_7 (0.47)	329.40 smp/s / 0.0000

Temperature Scaling Sensitivity

epochs = 10, KD without teacher annealing, full datasets

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
ledgar_KD_T2	1	9.00 / 0.000000	0.8421 / 0.000000	0.7821 / 0.000000	0.8590 / 0.000000	label_14 (1.00)	label_4 (0.00)	374.70 smp/s / 0.0000
unfair_tos_KD_T2	1	10.00 / 0.000000	0.0291 / 0.000000	0.7795 / 0.000000	0.7500 / 0.000000	label_5 (0.96)	label_4 (0.68)	382.30 smp/s / 0.0000
ledgar_KD_T4	1	9.00 / 0.000000	1.0475 / 0.000000	0.7580 / 0.000000	0.8546 / 0.000000	label_93 (0.99)	label_4 (0.00)	339.10 smp/s / 0.0000
unfair_tos_KD_T4	1	9.00 / 0.000000	0.0322 / 0.000000	0.7846 / 0.000000	0.7493 / 0.000000	label_5 (0.96)	label_0 (0.67)	341.30 smp/s / 0.0000
multi_task_kd_t2 (System)	1	10.00 / 0.000000	0.7279 / 0.000000	0.7272 / 0.000000	0.7623 / 0.000000	---	---	---
└─ LEDGAR Sub-Split	1	-	0.8382 / 0.000000	0.7790 / 0.000000	0.8595 / 0.000000	label_26 (0.99)	label_8 (0.00)	340.70 smp/s / 0.0000
└─ UNFAIR-ToS Sub-Split	1	-	0.0458 / 0.000000	0.6753 / 0.000000	0.6652 / 0.000000	label_6 (0.85)	label_7 (0.48)	335.80 smp/s / 0.0000
multi_task_kd_t4 (System)	1	10.00 / 0.000000	0.9154 / 0.000000	0.7108 / 0.000000	0.7591 / 0.000000	---	---	---
└─ LEDGAR Sub-Split	1	-	1.0552 / 0.000000	0.7534 / 0.000000	0.8530 / 0.000000	label_26 (0.99)	label_4 (0.00)	391.00 smp/s / 0.0000
└─ UNFAIR-ToS Sub-Split	1	-	0.0500 / 0.000000	0.6681 / 0.000000	0.6651 / 0.000000	label_6 (0.83)	label_7 (0.47)	391.40 smp/s / 0.0000
ledgar_L_m05_t2	3	10.00 / 0.000000	0.7804 / 0.000070	0.7866 / 0.000001	0.8601 / 0.000002	label_14 (1.00)	label_8 (0.00)	460.03 smp/s / 8.8133
unfair_tos_U_m05_t2	3	8.67 / 0.333333	0.0280 / 0.000002	0.7834 / 0.000429	0.7513 / 0.000170	label_5 (0.96)	label_0 (0.70) [S1] / label_0 (0.67) [S2] / label_2 (0.63) [S3]	459.20 smp/s / 0.4300
ledgar_L_m05_t4	3	10.00 / 0.000000	0.9138 / 0.000016	0.7678 / 0.000004	0.8565 / 0.000001	label_93 (0.99) [S1,S2] / label_26 (0.99) [S3]	label_8 (0.00)	463.83 smp/s / 11.3233
unfair_tos_U_m05_t4	3	9.33 / 0.333333	0.0316 / 0.000002	0.7630 / 0.000016	0.7404 / 0.000029	label_5 (0.96) [S1] / label_6 (0.94) [S2] / label_5 (0.92) [S3]	label_0 (0.67) [S1] / label_3 (0.67) [S2,S3]	461.73 smp/s / 11.9233
[MultiTask] mt_mix05_t2 (System)	3	10.00 / 0.000000	0.6742 / 0.000011	0.7263 / 0.000066	0.7534 / 0.000128	---	---	---
└─ LEDGAR Sub-Split	3	-	0.7756 / 0.000016	0.7827 / 0.000000	0.8593 / 0.000002	label_14 (1.00) [S1,S3] / label_26 (0.99) [S2]	label_8 (0.00)	453.63 smp/s / 34.1433
└─ UNFAIR-ToS Sub-Split	3	-	0.0474 / 0.000009	0.6699 / 0.000256	0.6475 / 0.000469	label_6 (0.88) [S1,S3] / label_6 (0.91) [S2]	label_7 (0.47) [S1] / label_7 (0.54) [S2] / label_7 (0.50) [S3]	450.27 smp/s / 53.6433
[MultiTask] mt_mix05_t4	3	10.00 / 0.000000	0.7807 / 0.000003	0.7199 / 0.000041	0.7517 / 0.000122	---	---	---

(System)								
└ LEDGAR Sub-Split	3	-	0.8987 / 0.000003	0.7671 / 0.000008	0.8563 / 0.000001	label_93 (0.99) [S1] / label_26 (0.99) [S2,S3]	label_4 (0.00) [S1] / label_8 (0.00) [S2,S3]	450.57 smp/s / 0.1633
└ UNFAIR-ToS Sub-Split	3	-	0.0506 / 0.000007	0.6726 / 0.000151	0.6470 / 0.000441	label_6 (0.90) [S1] / label_6 (0.94) [S2] / label_6 (0.87) [S3]	label_7 (0.52) [S1] / label_2 (0.53) [S2] / label_7 (0.47) [S3]	427.73 smp/s / 62.4633

Dynamic Teacher Annealing Performance

epochs = 10, full datasets, KD WEIGHT 1.0 → 0.0, T = 2

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
ledgar_KD_Annealing	3	8.33 / 1.333333	0.7368 / 0.000773	0.7802 / 0.000027	0.8563 / 0.000009	label_14 (1.00) [S1,S3] / label_93 (0.99) [S2]	label_8 (0.00)	358.73 smp/s / 355.8833
unfair_tos_KD_Annealing	3	9.00 / 1.000000	0.0270 / 0.000001	0.7792 / 0.000310	0.7510 / 0.000011	label_6 (0.94) [S1] / label_6 (0.97) [S2] / label_5 (0.96) [S3]	label_0 (0.68) [S1] / label_0 (0.67) [S2] / label_7 (0.63) [S3]	355.23 smp/s / 828.6233
multi_task_kd_annealing (System)	3	10.00 / 0.000000	0.5985 / 0.000002	0.7597 / 0.000062	0.7792 / 0.000113	---	---	---
└ LEDGAR Sub-Split	3	-	0.6896 / 0.000003	0.7861 / 0.000012	0.8594 / 0.000001	label_14 (1.00) [S1,S3] / label_80 (0.99) [S2]	label_8 (0.00)	376.13 smp/s / 1429.5733
└ UNFAIR-ToS Sub-Split	3	-	0.0349 / 0.000001	0.7332 / 0.000309	0.6990 / 0.000415	label_6 (0.93) [S1] / label_6 (0.97) [S2] / label_5 (0.88) [S3]	label_7 (0.61) [S1] / label_2 (0.59) [S2] / label_0 (0.60) [S3]	386.63 smp/s / 1781.3233

Resource-Constrained Scaling Trajectories

LEDGAR IS BALANCED TO MATCH UNFAIR-TOS DATASET SIZE, T = 2, EPOCHS INCREASED PROPORTIONALLY TO MATCH OBSERVED SAMPLES ON FULL DATASET

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
unfair_tos_Teacher_LR30	1	13.00 / 0.000000	0.0274 / 0.000000	0.7574 / 0.000000	0.7549 / 0.000000	label_5 (0.92)	label_7 (0.60)	20.70 smp/s / 0.0000
unfair_tos_Supervised_LR30	1	26.00 / 0.000000	0.0383 / 0.000000	0.6253 / 0.000000	0.6182 / 0.000000	label_7 (0.73)	label_3 (0.50)	346.90 smp/s / 0.0000
unfair_tos_KD_LR30	1	19.00 / 0.000000	0.0436 / 0.000000	0.6582 / 0.000000	0.6566 / 0.000000	label_5 (0.92)	label_7 (0.33)	380.70 smp/s / 0.0000
unfair_tos_Teacher_LR40	1	7.00 / 0.000000	0.0223 / 0.000000	0.8147 / 0.000000	0.8043 / 0.000000	label_6 (1.00)	label_7 (0.67)	17.80 smp/s / 0.0000
unfair_tos_Supervised_LR40	3	20.00 / 1.000000	0.0299 / 0.000000	0.7371 / 0.001139	0.7044 / 0.000379	label_6 (0.93) [S1] / label_5 (0.92) [S2] / label_5 (0.96) [S3]	label_2 (0.62) [S1] / label_7 (0.53) [S2] / label_2 (0.56) [S3]	356.50 smp/s / 527.7900
unfair_tos_KD_LR40	3	17.67 / 26.333333	0.0348 / 0.000001	0.7344 / 0.001406	0.7020 / 0.000248	label_5 (0.96) [S1] / label_5 (0.89) [S2] / label_5 (0.92) [S3]	label_0 (0.62) [S1] / label_7 (0.55) [S2] / label_0 (0.59) [S3]	343.53 smp/s / 138.8033
unfair_tos_Teacher_LR50	1	8.00 / 0.000000	0.0254 / 0.000000	0.8313 / 0.000000	0.7946 / 0.000000	label_6 (0.97)	label_2 (0.67)	17.70 smp/s / 0.0000
unfair_tos_Supervised_LR50	1	18.00 / 0.000000	0.0283 / 0.000000	0.7710 / 0.000000	0.7317 / 0.000000	label_6 (0.97)	label_2 (0.59)	338.10 smp/s / 0.0000
unfair_tos_KD_LR50	1	14.00 / 0.000000	0.0333 / 0.000000	0.7581 / 0.000000	0.7182 / 0.000000	label_6 (0.97)	label_2 (0.62)	347.70 smp/s / 0.0000
unfair_tos_Teacher_LR60	1	9.00 / 0.000000	0.0258 / 0.000000	0.8073 / 0.000000	0.7869 / 0.000000	label_5 (0.96)	label_2 (0.72)	18.20 smp/s / 0.0000
unfair_tos_Supervised_LR60	3	15.00 / 4.000000	0.0280 / 0.000000	0.7240 / 0.000285	0.6964 / 0.000207	label_5 (0.92) [S1] / label_5	label_2 (0.64) [S1] / label_2	345.27 smp/s / 16.0933

						(0.82) [S2] / label_5 (0.96) [S3]	(0.54) [S2] / label_7 (0.50) [S3]	
unfair_tos_KD_LR60	3	15.67 / 1.333333	0.0361 / 0.000002	0.7212 / 0.000470	0.7019 / 0.000178	label_6 (0.93) [S1] / label_5 (0.88) [S2] / label_5 (0.89) [S3]	label_4 (0.62) [S1] / label_2 (0.56) [S2] / label_3 (0.57) [S3]	345.53 smp/s / 324.0033
unfair_tos_Teacher_LR70	1	12.00 / 0.000000	0.0250 / 0.000000	0.8095 / 0.000000	0.7913 / 0.000000	label_6 (0.97)	label_7 (0.71)	17.80 smp/s / 0.0000
unfair_tos_Supervised_LR70	1	14.00 / 0.000000	0.0246 / 0.000000	0.7575 / 0.000000	0.7303 / 0.000000	label_6 (0.93)	label_4 (0.65)	339.50 smp/s / 0.0000
unfair_tos_KD_LR70	1	8.00 / 0.000000	0.0294 / 0.000000	0.7590 / 0.000000	0.7368 / 0.000000	label_6 (0.97)	label_3 (0.60)	355.40 smp/s / 0.0000
unfair_tos_Teacher_LR80	1	12.00 / 0.000000	0.0244 / 0.000000	0.8258 / 0.000000	0.8043 / 0.000000	label_6 (0.97)	label_0 (0.72)	21.00 smp/s / 0.0000
unfair_tos_Supervised_LR80	3	8.33 / 2.333333	0.0234 / 0.000000	0.7463 / 0.000234	0.7260 / 0.000319	label_5 (0.96) [S1] / label_6 (0.97) [S2] / label_5 (0.92) [S3]	label_2 (0.66) [S1] / label_7 (0.53) [S2] / label_2 (0.65) [S3]	362.00 smp/s / 1646.9100
unfair_tos_KD_LR80	3	13.00 / 0.000000	0.0309 / 0.000003	0.7717 / 0.000331	0.7484 / 0.000196	label_6 (0.94) [S1,S2] / label_5 (0.96) [S3]	label_3 (0.67) [S1] / label_2 (0.68) [S2] / label_7 (0.63) [S3]	336.57 smp/s / 104.7033
unfair_tos_Teacher_LR90	1	5.00 / 0.000000	0.0243 / 0.000000	0.8212 / 0.000000	0.8011 / 0.000000	label_6 (1.00)	label_2 (0.69)	21.20 smp/s / 0.0000
unfair_tos_Supervised_LR90	1	8.00 / 0.000000	0.0257 / 0.000000	0.7675 / 0.000000	0.7277 / 0.000000	label_6 (0.94)	label_0 (0.65)	357.70 smp/s / 0.0000
unfair_tos_KD_LR90	1	11.00 / 0.000000	0.0295 / 0.000000	0.7765 / 0.000000	0.7500 / 0.000000	label_5 (0.96)	label_2 (0.67)	340.20 smp/s / 0.0000

LEDGAR Single-Task Low-Resource Metrics

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
ledgar_Teacher_LR3	1	31.00 / 0.000000	1.3969 / 0.000000	0.6805 / 0.000000	0.7973 / 0.000000	label_16 (0.99)	label_8 (0.00)	18.70 smp/s / 0.0000
ledgar_Supervised_LR3	1	25.00 / 0.000000	1.1673 / 0.000000	0.5878 / 0.000000	0.7301 / 0.000000	label_26 (0.98)	label_4 (0.00)	404.40 smp/s / 0.0000
ledgar_KD_LR3	1	32.00 / 0.000000	1.4157 / 0.000000	0.5583 / 0.000000	0.7270 / 0.000000	label_26 (0.98)	label_4 (0.00)	318.20 smp/s / 0.0000
ledgar_Teacher_LR4	1	15.00 / 0.000000	1.2421 / 0.000000	0.6936 / 0.000000	0.8033 / 0.000000	label_80 (0.99)	label_8 (0.00)	18.30 smp/s / 0.0000
ledgar_Supervised_LR4	3	23.67 / 2.333333	1.0497 / 0.000449	0.6084 / 0.000030	0.7513 / 0.000038	label_26 (0.98)	label_4 (0.00) [S1] / label_8 (0.00) [S2, S3]	338.00 smp/s / 153.8100
ledgar_KD_LR4	3	24.67 / 0.333333	1.2883 / 0.000724	0.5699 / 0.000070	0.7427 / 0.000028	label_26 (0.98)	label_1 (0.00)	328.83 smp/s / 1707.0233
ledgar_Teacher_LR5	1	17.00 / 0.000000	1.2104 / 0.000000	0.7109 / 0.000000	0.8137 / 0.000000	label_80 (0.99)	label_8 (0.00)	18.10 smp/s / 0.0000
ledgar_Supervised_LR5	1	20.00 / 0.000000	0.9792 / 0.000000	0.6259 / 0.000000	0.7642 / 0.000000	label_26 (0.98)	label_8 (0.00)	345.60 smp/s / 0.0000
ledgar_KD_LR5	1	20.00 / 0.000000	1.2507 / 0.000000	0.5773 / 0.000000	0.7519 / 0.000000	label_26 (0.98)	label_1 (0.00)	327.60 smp/s / 0.0000
ledgar_Teacher_LR6	1	14.00 / 0.000000	1.1902 / 0.000000	0.7094 / 0.000000	0.8115 / 0.000000	label_16 (0.99)	label_8 (0.00)	18.40 smp/s / 0.0000
ledgar_Supervised_LR6	3	16.67 / 0.333333	0.9657 / 0.000439	0.6246 / 0.000109	0.7659 / 0.000036	label_26 (0.98)	label_4 (0.00) [S1] / label_1 (0.00) [S2] / label_8 (0.00) [S3]	343.27 smp/s / 41.9233
ledgar_KD_LR6	3	16.67 / 0.333333	1.2292 / 0.000221	0.5830 / 0.000042	0.7565 / 0.000020	label_26 (0.98)	label_3 (0.00)	340.17 smp/s / 624.0033
ledgar_Teacher_LR7	1	12.00 / 0.000000	1.1660 / 0.000000	0.7215 / 0.000000	0.8169 / 0.000000	label_43 (0.99)	label_8 (0.00)	18.00 smp/s / 0.0000

ledgar_Supervised_LR7	1	13.00 / 0.000000	0.9415 / 0.000000	0.6274 / 0.000000	0.7675 / 0.000000	label_26 (0.99)	label_4 (0.00)	341.90 smp/s / 0.0000
ledgar_KD_LR7	1	14.00 / 0.000000	1.2054 / 0.000000	0.5793 / 0.000000	0.7546 / 0.000000	label_26 (0.98)	label_1 (0.00)	325.60 smp/s / 0.0000
ledgar_Teacher_LR8	1	13.00 / 0.000000	1.0806 / 0.000000	0.7304 / 0.000000	0.8218 / 0.000000	label_26 (0.99)	label_4 (0.10)	22.30 smp/s / 0.0000
ledgar_Supervised_LR8	3	12.00 / 0.000000	0.9243 / 0.000529	0.6374 / 0.000133	0.7748 / 0.000032	label_26 (0.99) [S1] / label_26 (0.98) [S2,S3]	label_4 (0.00) [S1] / label_8 (0.00) [S2,S3]	342.57 smp/s / 49.4633
ledgar_KD_LR8	3	12.67 / 0.333333	1.1954 / 0.000259	0.5896 / 0.000020	0.7619 / 0.000013	label_26 (0.98)	label_3 (0.00) [S1,S2] / label_1 (0.00) [S3]	343.80 smp/s / 87.6100
ledgar_Teacher_LR9	1	9.00 / 0.000000	0.9865 / 0.000000	0.7389 / 0.000000	0.8243 / 0.000000	label_16 (0.99)	label_4 (0.08)	20.20 smp/s / 0.0000
ledgar_Supervised_LR9	1	11.00 / 0.000000	0.9177 / 0.000000	0.6283 / 0.000000	0.7715 / 0.000000	label_26 (0.99)	label_4 (0.00)	334.20 smp/s / 0.0000
ledgar_KD_LR9	1	11.00 / 0.000000	1.1814 / 0.000000	0.5838 / 0.000000	0.7578 / 0.000000	label_26 (0.98)	label_1 (0.00)	356.10 smp/s / 0.0000
ledgar_Teacher_LR10	1	9.00 / 0.000000	0.9602 / 0.000000	0.7422 / 0.000000	0.8273 / 0.000000	label_15 (0.99)	label_8 (0.00)	21.80 smp/s / 0.0000
ledgar_Baseline_LR10	3	10.00 / 0.000000	0.9058 / 0.000309	0.6333 / 0.000083	0.7769 / 0.000030	label_26 (0.98)	label_4 (0.00)	335.10 smp/s / 112.2700
ledgar_KD_Student_LR10	3	10.00 / 0.000000	1.1440 / 0.000169	0.5860 / 0.000065	0.7619 / 0.000013	label_26 (0.98)	label_1 (0.00)	336.10 smp/s / 3.0000

Multi-Task Low-Resource Metrics

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
MT_Supervised_LR30_3 (System)	1	20.00 / 0.000000	0.9937 / 0.000000	0.5785 / 0.000000	0.6619 / 0.000000	---	---	---
└─ LEDGAR Sub-Split	1	-	1.1481 / 0.000000	0.5832 / 0.000000	0.7308 / 0.000000	label_26 (0.98)	label_4 (0.00)	339.20 smp/s / 0.0000
└─ UNFAIR-ToS Sub-Split	1	-	0.0381 / 0.000000	0.5738 / 0.000000	0.5930 / 0.000000	label_5 (0.88)	label_7 (0.25)	341.30 smp/s / 0.0000
MT_KD_LR30_3 (System)	1	32.00 / 0.000000	1.2237 / 0.000000	0.5642 / 0.000000	0.6694 / 0.000000	---	---	---
└─ LEDGAR Sub-Split	1	-	1.4132 / 0.000000	0.5585 / 0.000000	0.7335 / 0.000000	label_26 (0.97)	label_1 (0.00)	336.30 smp/s / 0.0000
└─ UNFAIR-ToS Sub-Split	1	-	0.0512 / 0.000000	0.5699 / 0.000000	0.6053 / 0.000000	label_5 (0.85)	label_7 (0.12)	320.50 smp/s / 0.0000
MT_Supervised_LR40_4 (System)	3	17.67 / 9.333333	0.9410 / 0.000661	0.6327 / 0.000168	0.7045 / 0.000047	---	---	---
└─ LEDGAR Sub-Split	3	-	1.0878 / 0.000899	0.5970 / 0.000151	0.7454 / 0.000051	label_26 (0.98)	label_4 (0.00) [S1] / label_5 (0.00) [S2] / label_8 (0.00) [S3]	360.03 smp/s / 961.8833
└─ UNFAIR-ToS Sub-Split	3	-	0.0328 / 0.000002	0.6684 / 0.000209	0.6635 / 0.000051	label_5 (0.88) [S1,S2] / label_5 (0.80) [S3]	label_2 (0.58) [S1] / label_7 (0.50) [S2] / label_7 (0.56) [S3]	361.43 smp/s / 74.6033
MT_KD_LR40_4 (System)	3	22.67 / 4.333333	1.1357 / 0.000002	0.6232 / 0.000017	0.7069 / 0.000088	---	---	---
└─ LEDGAR Sub-Split	3	-	1.3135 / 0.000002	0.5585 / 0.000038	0.7383 / 0.000016	label_26 (0.98)	label_1 (0.00) [S1,S2] / label_5 (0.00) [S3]	358.23 smp/s / 290.8233
└─ UNFAIR-ToS Sub-Split	3	-	0.0358 / 0.000001	0.6880 / 0.000083	0.6754 / 0.000219	label_5 (0.81) [S1,S2] / label_6 (0.81) [S3]	label_7 (0.62) [S1] / label_0 (0.60) [S2] / label_4 (0.41) [S3]	370.20 smp/s / 796.8900
MT_Supervised_LR50_5 (System)	1	19.00 / 0.000000	0.8594 / 0.000000	0.6461 / 0.000000	0.7063 / 0.000000	---	---	---
└─ LEDGAR Sub-Split	1	-	0.9932 / 0.000000	0.6299 / 0.000000	0.7621 / 0.000000	label_26 (0.98)	label_1 (0.00)	368.80 smp/s / 0.0000

└ UNFAIR-ToS Sub-Split	1	-	0.0320 / 0.000000	0.6624 / 0.000000	0.6505 / 0.000000	label_5 (0.83)	label_4 (0.55)	339.20 smp/s / 0.0000
MT_KD_LR50_5 (System)	1	18.00 / 0.000000	1.1183 / 0.000000	0.6023 / 0.000000	0.6927 / 0.000000	---	---	---
└ LEDGAR Sub-Split	1	-	1.2930 / 0.000000	0.5708 / 0.000000	0.7485 / 0.000000	label_26 (0.98)	label_1 (0.00)	383.10 smp/s / 0.0000
└ UNFAIR-ToS Sub-Split	1	-	0.0372 / 0.000000	0.6338 / 0.000000	0.6369 / 0.000000	label_5 (0.83)	label_4 (0.45)	346.30 smp/s / 0.0000
MT_Supervised_LR60_6 (System)	3	15.33 / 4.333333	0.8452 / 0.000102	0.6656 / 0.000035	0.7317 / 0.000056	---	---	---
└ LEDGAR Sub-Split	3	-	0.9771 / 0.000143	0.6284 / 0.000042	0.7668 / 0.000015	label_26 (0.98)	label_4 (0.00) [S1,S2] / label_8 (0.00) [S3]	368.90 smp/s / 518.2300
└ UNFAIR-ToS Sub-Split	3	-	0.0292 / 0.000005	0.7029 / 0.000146	0.6966 / 0.000289	label_5 (0.88) [S1] / label_5 (0.83) [S2] / label_5 (0.92) [S3]	label_3 (0.55) [S1] / label_7 (0.59) [S2] / label_7 (0.55) [S3]	365.37 smp/s / 688.1633
MT_KD_LR60_6 (System)	3	14.33 / 5.333333	1.0968 / 0.000629	0.6360 / 0.000022	0.7152 / 0.000022	---	---	---
└ LEDGAR Sub-Split	3	-	1.2682 / 0.000859	0.5694 / 0.000104	0.7471 / 0.000023	label_26 (0.98) [S1,S2] / label_26 (0.97) [S3]	label_1 (0.00) [S1] / label_3 (0.00) [S2,S3]	347.33 smp/s / 7.1233
└ UNFAIR-ToS Sub-Split	3	-	0.0365 / 0.000001	0.7026 / 0.000087	0.6833 / 0.000174	label_5 (0.88) [S1,S2] / label_5 (0.85) [S3]	label_4 (0.59) [S1] / label_4 (0.46) [S2] / label_4 (0.51) [S3]	361.97 smp/s / 848.4233
MT_Supervised_LR70_7 (System)	1	14.00 / 0.000000	0.8168 / 0.000000	0.6831 / 0.000000	0.7416 / 0.000000	---	---	---
└ LEDGAR Sub-Split	1	-	0.9443 / 0.000000	0.6359 / 0.000000	0.7742 / 0.000000	label_26 (0.98)	label_4 (0.00)	335.00 smp/s / 0.0000
└ UNFAIR-ToS Sub-Split	1	-	0.0277 / 0.000000	0.7304 / 0.000000	0.7090 / 0.000000	label_5 (0.88)	label_3 (0.57)	413.30 smp/s / 0.0000
MT_KD_LR70_7 (System)	1	13.00 / 0.000000	1.0772 / 0.000000	0.6426 / 0.000000	0.7300 / 0.000000	---	---	---
└ LEDGAR Sub-Split	1	-	1.2458 / 0.000000	0.5689 / 0.000000	0.7516 / 0.000000	label_26 (0.98)	label_3 (0.00)	345.60 smp/s / 0.0000
└ UNFAIR-ToS Sub-Split	1	-	0.0336 / 0.000000	0.7162 / 0.000000	0.7084 / 0.000000	label_5 (0.88)	label_3 (0.64)	334.20 smp/s / 0.0000
MT_Supervised_LR80_8 (System)	3	11.67 / 0.333333	0.8053 / 0.000147	0.6834 / 0.000006	0.7445 / 0.000006	---	---	---
└ LEDGAR Sub-Split	3	-	0.9311 / 0.000196	0.6357 / 0.000009	0.7744 / 0.000016	label_26 (0.98) [S1,S3] / label_15 (0.98) [S2]	label_4 (0.00) [S1,S2] / label_8 (0.00) [S3]	351.03 smp/s / 739.0633
└ UNFAIR-ToS Sub-Split	3	-	0.0270 / 0.000004	0.7311 / 0.000006	0.7146 / 0.000025	label_5 (0.88) [S1] / label_6 (0.90) [S2] / label_5 (0.83) [S3]	label_4 (0.59) [S1] / label_2 (0.59) [S2] / label_3 (0.65) [S3]	349.27 smp/s / 355.2933
MT_KD_LR80_8 (System)	3	12.67 / 0.333333	1.0634 / 0.000131	0.6506 / 0.000056	0.7320 / 0.000050	---	---	---
└ LEDGAR Sub-Split	3	-	1.2303 / 0.000182	0.5782 / 0.000004	0.7553 / 0.000001	label_26 (0.98) [S1,S2] / label_26 (0.97) [S3]	label_1 (0.00)	361.50 smp/s / 418.6800
└ UNFAIR-ToS Sub-Split	3	-	0.0310 / 0.000002	0.7230 / 0.000193	0.7087 / 0.000229	label_5 (0.88) [S1,S2] / label_5 (0.92) [S3]	label_4 (0.56) [S1] / label_2 (0.60) [S2] / label_4 (0.65) [S3]	365.33 smp/s / 379.8533
MT_Supervised_LR90_9 (System)	1	10.00 / 0.000000	0.7917 / 0.000000	0.6760 / 0.000000	0.7380 / 0.000000	---	---	---
└ LEDGAR Sub-Split	1	-	0.9153 / 0.000000	0.6410 / 0.000000	0.7788 / 0.000000	label_26 (0.98)	label_4 (0.00)	336.60 smp/s / 0.0000
└ UNFAIR-ToS Sub-Split	1	-	0.0269 / 0.000000	0.7109 / 0.000000	0.6973 / 0.000000	label_5 (0.88)	label_2 (0.63)	340.60 smp/s / 0.0000

MT_KD_LR90_9 (System)	1	11.00 / 0.000000	1.0316 / 0.000000	0.6424 / 0.000000	0.7302 / 0.000000	---	---	---
└─ LEDGAR Sub-Split	1	-	1.1934 / 0.000000	0.5728 / 0.000000	0.7552 / 0.000000	label_64 (0.98)	label_1 (0.00)	379.20 smp/s / 0.0000
└─ UNFAIR-ToS Sub-Split	1	-	0.0304 / 0.000000	0.7119 / 0.000000	0.7052 / 0.000000	label_5 (0.88)	label_4 (0.62)	376.90 smp/s / 0.0000
multi_task_super_lr100_10 (System)	3	9.67 / 0.333333	0.7792 / 0.000022	0.6946 / 0.000029	0.7562 / 0.000027	---	---	---
└─ LEDGAR Sub-Split	3	-	0.9010 / 0.000030	0.6385 / 0.000002	0.7791 / 0.000001	label_26 (0.98)	label_4 (0.00) [S1,S2] / label_8 (0.00) [S3]	402.47 smp/s / 412.3033
└─ UNFAIR-ToS Sub-Split	3	-	0.0259 / 0.000002	0.7507 / 0.000087	0.7333 / 0.000126	label_5 (0.88) [S1,S2] / label_5 (0.92) [S3]	label_3 (0.62) [S1] / label_2 (0.64) [S2] / label_7 (0.67) [S3]	396.73 smp/s / 235.0533
multi_task_kd_lr100_10 (System)	3	10.00 / 0.000000	1.0128 / 0.000027	0.6555 / 0.000064	0.7383 / 0.000068	---	---	---
└─ LEDGAR Sub-Split	3	-	1.1715 / 0.000038	0.5775 / 0.000028	0.7566 / 0.000002	label_26 (0.97) [S1,S3] / label_26 (0.98) [S2]	label_1 (0.00)	350.83 smp/s / 84.2833
└─ UNFAIR-ToS Sub-Split	3	-	0.0303 / 0.000001	0.7336 / 0.000123	0.7200 / 0.000232	label_5 (0.92) [S1,S3] / label_5 (0.88) [S2]	label_7 (0.56) [S1] / label_2 (0.63) [S2] / label_7 (0.63) [S3]	326.50 smp/s / 268.0300

Low resource experiments with Teacher annealing as distillation method

- Temperatures = {2, 4}, Three seeds

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
ledgar_L_ta40_4_t2	3	25.00 / 0.000000	1.1294 / 0.000363	0.6034 / 0.000002	0.7488 / 0.000005	label_26 (0.98)	label_4 (0.00) [S1] / label_1 (0.00) [S2,S3]	462.30 smp/s / 0.1300
unfair_tos_U_ta40_4_t2	3	19.67 / 1.333333	0.0343 / 0.000002	0.7552 / 0.000280	0.7137 / 0.000235	label_5 (0.96) [S1] / label_5 (0.89) [S2] / label_5 (0.92) [S3]	label_0 (0.61) [S1] / label_0 (0.63) [S2] / label_0 (0.60) [S3]	459.93 smp/s / 0.2533
ledgar_L_ta40_4_t4	3	25.00 / 0.000000	1.1292 / 0.000164	0.6037 / 0.000003	0.7503 / 0.000002	label_26 (0.98)	label_4 (0.00) [S1] / label_1 (0.00) [S2,S3]	459.77 smp/s / 1.0033
unfair_tos_U_ta40_4_t4	3	23.33 / 0.333333	0.0362 / 0.000003	0.7566 / 0.000087	0.7246 / 0.000080	label_5 (0.96) [S1] / label_5 (0.92) [S2,S3]	label_0 (0.62) [S1] / label_2 (0.66) [S2] / label_2 (0.63) [S3]	460.57 smp/s / 2.3433
[MultiTask] mt_ta40_4_t2 (System)	3	24.33 / 0.333333	0.9957 / 0.000007	0.6412 / 0.000021	0.7110 / 0.000013	---	---	---
└─ LEDGAR Sub-Split	3	-	1.1511 / 0.000009	0.5933 / 0.000016	0.7461 / 0.000004	label_26 (0.97) [S1] / label_26 (0.98) [S2,S3]	label_4 (0.00) [S1] / label_1 (0.00) [S2] / label_8 (0.00) [S3]	454.70 smp/s / 3.0100
└─ UNFAIR-ToS Sub-Split	3	-	0.0341 / 0.000002	0.6892 / 0.000170	0.6759 / 0.000028	label_5 (0.79) [S1] / label_5 (0.88) [S2] / label_6 (0.81) [S3]	label_3 (0.62) [S1] / label_4 (0.56) [S2] / label_4 (0.54) [S3]	427.50 smp/s / 1.1200
[MultiTask] mt_ta40_4_t4 (System)	3	24.33 / 0.333333	1.0022 / 0.000001	0.6378 / 0.000011	0.7081 / 0.000044	---	---	---
└─ LEDGAR Sub-Split	3	-	1.1584 / 0.000002	0.5821 / 0.000010	0.7402 / 0.000008	label_26 (0.98)	label_1 (0.00) [S1,S2] / label_5 (0.00) [S3]	469.60 smp/s / 20.2300
└─ UNFAIR-ToS Sub-Split	3	-	0.0354 / 0.000003	0.6935 / 0.000016	0.6761 / 0.000125	label_5 (0.85) [S1] / label_5 (0.88) [S2] / label_7 (0.80) [S3]	label_4 (0.60) [S1] / label_4 (0.54) [S2] / label_4 (0.53) [S3]	441.07 smp/s / 8.1733

Model / Task Sub-Split	Seeds (N)	Epoch (μ/σ^2)	Loss (μ/σ^2)	Macro-F1 (μ/σ^2)	Micro-F1 (μ/σ^2)	Best Label (F1)	Worst Label (F1)	Throughput (μ/σ^2)
[MultiTask] mt_ta50_5_t2 (System)	3	19.67 / 0.333333	0.9548 / 0.000039	0.6377 / 0.000170	0.7089 / 0.000059	---	---	---
└─ LEDGAR Sub-Split	3	-	1.1035 / 0.000054	0.5995 / 0.000036	0.7523 / 0.000017	label_26 (0.98)	label_1 (0.00) [S1,S2] / label_8 (0.00) [S3]	325.50 smp/s / 32.8300
└─ UNFAIR-ToS Sub-Split	3	-	0.0344 / 0.000001	0.6759 / 0.000957	0.6655 / 0.000381	label_5 (0.88) [S1] / label_6 (0.90) [S2] / label_5 (0.83) [S3]	label_7 (0.53) [S1] / label_2 (0.46) [S2] / label_2 (0.58) [S3]	325.30 smp/s / 21.0700
[MultiTask] mt_ta50_5_t4 (System)	3	19.67 / 0.333333	0.9628 / 0.000199	0.6404 / 0.000266	0.7098 / 0.000051	---	---	---
└─ LEDGAR Sub-Split	3	-	1.1127 / 0.000264	0.5956 / 0.000019	0.7500 / 0.000012	label_26 (0.98) [S1,S2] / label_26 (0.97) [S3]	label_1 (0.00) [S1,S2] / label_8 (0.00) [S3]	338.87 smp/s / 30.0133
└─ UNFAIR-ToS Sub-Split	3	-	0.0357 / 0.000000	0.6852 / 0.000957	0.6697 / 0.000174	label_5 (0.88) [S1,S3] / label_5 (0.87) [S2]	label_7 (0.53) [S1,S3] / label_2 (0.54) [S2]	337.57 smp/s / 47.2233
[MultiTask] mt_ta70_7_t2 (System)	3	13.33 / 1.333333	0.9125 / 0.000133	0.6668 / 0.000006	0.7375 / 0.000001	---	---	---
└─ LEDGAR Sub-Split	3	-	1.0552 / 0.000178	0.6068 / 0.000110	0.7615 / 0.000004	label_26 (0.98)	label_3 (0.00) [S1] / label_4 (0.00) [S2] / label_5 (0.00) [S3]	340.73 smp/s / 91.7233
└─ UNFAIR-ToS Sub-Split	3	-	0.0294 / 0.000000	0.7268 / 0.000068	0.7135 / 0.000004	label_5 (0.85) [S1] / label_6 (0.90) [S2] / label_5 (0.83) [S3]	label_3 (0.64) [S1] / label_3 (0.59) [S2] / label_3 (0.62) [S3]	346.07 smp/s / 259.5433
[MultiTask] mt_ta70_7_t4 (System)	3	13.67 / 0.333333	0.9124 / 0.000154	0.6697 / 0.000022	0.7398 / 0.000010	---	---	---
└─ LEDGAR Sub-Split	3	-	1.0548 / 0.000208	0.6012 / 0.000039	0.7581 / 0.000015	label_26 (0.97) [S1] / label_26 (0.98) [S2,S3]	label_4 (0.00) [S1,S2] / label_8 (0.00) [S3]	339.20 smp/s / 122.4700
└─ UNFAIR-ToS Sub-Split	3	-	0.0316 / 0.000000	0.7382 / 0.000016	0.7214 / 0.000011	label_5 (0.88) [S1] / label_5 (0.92) [S2] / label_5 (0.83) [S3]	label_3 (0.65) [S1] / label_3 (0.61) [S2] / label_4 (0.67) [S3]	334.67 smp/s / 157.6033
[MultiTask] mt_ta100_10_t2 (System)	3	10.00 / 0.000000	0.8399 / 0.000019	0.6791 / 0.000040	0.7461 / 0.000017	---	---	---
└─ LEDGAR Sub-Split	3	-	0.9712 / 0.000027	0.6187 / 0.000019	0.7702 / 0.000006	label_26 (0.98)	label_1 (0.00) [S1,S2] / label_8 (0.00) [S3]	332.23 smp/s / 0.6533
└─ UNFAIR-ToS Sub-Split	3	-	0.0277 / 0.000001	0.7396 / 0.000087	0.7220 / 0.000034	label_5 (0.88) [S1,S2] / label_5 (0.92) [S3]	label_3 (0.62) [S1] / label_2 (0.61) [S2] / label_7 (0.60) [S3]	330.80 smp/s / 214.8100
[MultiTask] mt_ta100_10_t4 (System)	3	9.67 / 0.333333	0.8329 / 0.000073	0.6795 / 0.000034	0.7449 / 0.000054	---	---	---
└─ LEDGAR Sub-Split	3	-	0.9627 / 0.000098	0.6168 / 0.000015	0.7694 / 0.000012	label_26 (0.98)	label_1 (0.00) [S1] / label_4 (0.00) [S2] / label_8 (0.00) [S3]	335.13 smp/s / 9.9033
└─ UNFAIR-ToS Sub-Split	3	-	0.0302 / 0.000001	0.7423 / 0.000064	0.7203 / 0.000139	label_5 (0.92) [S1,S2] / label_5 (0.88) [S3]	label_7 (0.64) [S1] / label_2 (0.62) [S2] / label_7 (0.63) [S3]	335.20 smp/s / 21.2700